Daniela Caraeni

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Summary

Aerospace-focused mechanical engineer with 7+ years of experience in computational fluid dynamics (CFD), fluid-structure interaction (FSI), and multiphase flow simulation. Expertise in ANSYS, OpenFOAM, and high-fidelity modeling of complex dynamic systems. Proven ability to translate advanced simulations into actionable engineering insights, with strong communication skills for technical and cross-functional audiences.

Core Competencies

Computational Fluid Dynamics | Fluid–Structure Interaction | Multiphase Flow | ANSYS Fluent and CFX | OpenFOAM | High-Performance Computing | Numerical Methods and Modeling | Data Analysis | MATLAB | Python | C/C++ | FORTRAN | English (Fluent) | French (Intermediate)

Education

Doctor of Philosophy, Mechanical and Industrial Engineering — **Amherst, MA**
University of Massachusetts Amherst — *ABD*

Master of Engineering, Mechanical Engineering — **Montreal, QC**
McGill University — *Dec 2019*

Bachelor of Engineering, Mechanical Engineering — **Montreal, QC**
McGill University — *May 2017*

Experience

Graduate Research Assistant in the UMASS FSI Lab — **Amherst, MA**
University of Massachusetts — *Sept 2020 – present*

- Designed and executed high-fidelity CFD simulations for coronary arteries using ANSYS Fluent and POD to analyze hemodynamic behavior before and after stent placement.
- Modeled dynamic interactions of multiple free-moving cylinders in tandem and side-by-side configurations to understand vortex-induced and wake-induced vibrations.
- Simulated flexible shell deformations in ANSYS Workbench using coupled FSI solvers to evaluate structural response under unsteady flow conditions.
- Led undergraduate mechanics lab instruction, managing lab content and assessments.

Graduate Student Researcher in the McGill CFD Lab — **Montreal, QC**
McGill University — *Sept 2017 – Dec 2019*

- Developed a multiphase FSI solver using XFEM and level set methods to simulate solidification in complex geometries—a technique applicable to ice accretion on aircraft components.
- Integrated Brinkman penalization and Stefan energy models into a CFD framework to model thermal phase change at solid-fluid boundaries.
- Presented collaborative research at AIAA SciTech and disseminated findings through peer-reviewed publication in *Finite Elements in Analysis and Design*, influencing ongoing XFEM solver developments.

Software Intern — **Wilder, VT**
Concepts NREC — *May 2016 – August 2016*

- Optimized CFD post-processing workflows for turbomachinery design, implementing Bézier interpolation and profile-smoothing algorithms to improve turbine blade loss analysis.
- Conducted and interpreted AxCent® performance simulations to support virtual test rigs for axial turbine blade development.
- Audited and corrected turbomachinery airfoil libraries (CIAM, Deytch profiles), contributing to benchmarking and performance validation of gas turbine systems.

Selected Projects

Computational Fluid Dynamics *2023*
- Conducted a numerical investigation of symmetric and asymmetric vortex shedding behind a 1-DOF cylinder undergoing vortex-induced vibrations
- Leveraged high-performance computing resources to run parallel simulations on the Unity cluster, significantly enhancing computational efficiency

Nonlinear Dynamics *2022*
- Analyzed energy harvesting potential of chaotic pendulums by applying nonlinear dynamics to single and double pendulum systems across periodic and chaotic regimes
- Identified key dynamic behaviors to optimize energy capture, contributing to improved understanding of nonlinear energy harvesting mechanisms

Fluid-Structure Interactions *2020*
- Investigated paragliding as a fluid-structure interaction system, emphasizing aerodynamic and structural coupling
- Analyzed instability onset driven by pilot center of gravity, line stiffness, and pilot weight to understand flight dynamics and control limitations

Presentations

Graduate Research Symposium *Amherst, Massachusetts, 2025*
- Presented poster Dynamic Motion: Exploring Free-Oscillating Cylinders in Diverse Formations to an interdisciplinary engineering audience; awarded Runner-Up for poster quality and content by an independent faculty panel.
- Delivered a 3-minute "sound bite" talk summarizing research to the full symposium audience of graduate and undergraduate engineering students.

APS Division of Fluid Dynamics Meeting *USA, 2021–2024*
- Delivered four consecutive presentations across multiple research topics in fluid–structure interaction and computational fluid dynamics to audiences of $\approx$ 50–100 attendees.
- Selected for DFD-Interact series; presented a 2-minute talk and 1-hour poster session. Also contributed two works to the Gallery of Fluid Motion:
  - Poster showing flow transitions from jet formation to vortex shedding in kirigami and rigid sheets.
  - Video showing collective motion and wake reconfiguration of free-oscillating cylinders in various arrays.

Professional & Community Engagement

Dean's Advisory Group *2024–2025*
- Provided input on academic program development and student experience initiatives, representing graduate student perspectives to the Dean's Office.
- Collaborated with faculty and administrators to review and recommend policies aimed at improving research support and campus resources.

Graduate Women in STEM *2020–2023*
- Participated in outreach events and mentoring programs to support the retention and success of women pursuing graduate studies in STEM fields.
- Assisted in organizing workshops for attracting 50+ school aged attendees from across the Pioneer Valley.

Selected Publications

- D. Caraeni, H. Gong, Y. Modarres-Sadeghi (2026). Collective Behavior of Clusters of Free-to-Move Cylinders in the Wake of a Fixed Cylinder. *Journal of Fluid Mechanics*. 1035, A31.
- D. Caraeni. H. Gong, Y. Modarres-Sadeghi (2026). Modeling the influence of drafting formations in running on the experienced drag force using fixed and free-to-move bluff bodies. *Journal of Fluids and Structures*. 148, 104715.
- A. Lotfi, D. Caraeni, O. Haider, A. Pervaiz, Y. Modarres-Sadeghi (2025). Computational Fluid Dynamics Model Utilizing Proper Orthogonal Decomposition to Assess Coronary Physiology and Wall Shear Stress. *Computers in Biology and Medicine*. 188, 109840.
- D. Caraeni, V. Casseau, W. G. Habashi, "Fluid-Structure Interaction: Extended-FEM Approach to Solidification", *Finite Elements in Analysis and Design*, 177 (2020): 103425.